

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			Ammeter in series (1) Voltmeter shown in parallel with the thermistor (1) N.B. correct symbol shown in a circle	2			2		2
	(b)			6 points plotted correctly to within <1 small square tolerance (1) 5 points or less plotted correctly to within <1 small square tolerance (0) Curve of best fit between 0-125 °C <1 small square tolerance from plotted points (1)		2		2	2	2
	(c)	(i)		Total resistance = 40 + 40 = 80 [Ω] (1) so $I = \frac{12}{80} = 0.15$ [A] (1) Therefore $V = 0.15 \times 40 = 6$ [V] (1) Alternative: Voltage shared equally (1) so $V = \frac{12}{2}$ (1) = 6 [V] (1)		3		3	3	3
		(ii)		As temperature increases, {the resistance of the thermistor decreases/ the resistance shown by the {graph / table} decreases (1) Total [series] resistance decreases, [so current increases] (1) [so agree]			2	2		2

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	(d)	(i)		$I = \frac{3}{12} = 0.25 \text{ [A]} \text{ (1)}$ Rearrangement: $R = \frac{12}{0.25 \text{ ecf}} \text{ (1)}$ $= 48 \text{ [\Omega]} \text{ (1)}$ Alternative: $I = \frac{3}{12} = 0.25 \text{ [A]} \text{ (1)}$ Rearrangement: $R = \frac{P}{I^2} = \frac{3}{0.25^2 \text{ ecf}}$ $= 48 \text{ [\Omega]} \text{ (1)}$		3		3	3	3
		(ii)		[Total resistance decreases] because the {lamp / thermistor} is now in parallel or there is another path for the current (1) So the parallel resistance is lower [than the thermistor resistance on its own so the total resistance is lower] (1)	1	1		2		2
				Question 7 total	3	9	2	14	8	14